

Shreya Sunil Kale

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PROFESSIONAL SUMMARY

Software Engineer and M.S. in Computer Science student at Arizona State University with 2+ years of experience building Agentic AI and AI-driven systems using modern cloud and distributed technologies, with expertise in cybersecurity and threat research.

EDUCATION

Arizona State University

Masters of Science in Computer Science

Tempe, Arizona

Aug 2025 - May 2027

- **Coursework:** Statistical Machine Learning, Knowledge Representation & Reasoning, Engr Blockchain Applications, Data Mining

Savitribai Phule Pune University

Bachelor of Engineering in Computer Engineering

Pune, India

Aug 2019 - Apr 2023

- **Coursework:** Data Structures and Algorithm, Software Development, Cloud Computing, Object Oriented Programming, System Design, Operating System

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, TypeScript, Go Lang

ML & Cyber: Django, Flask, OpenCV, TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, SIEM, MITRE ATT&CK

Web Development : Angular, Node.js, Spring Boot, React.js, Express.js, Rest API

Big Data & Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB), Apache Spark, Kafka

Tools: Git, Gitlab, GitHub, CI/CD, Jenkins, Docker, Kubernetes, Prometheus, JIRA, Wireshark, x64dbg, OllyDbg, DnSpy

Cloud Platforms: AWS(EKS, S3, EC2, SQS, ELB, Lambda, DynamoDB, CloudFront, CloudFormation), Google Cloud Platform

WORK EXPERIENCE

LTIMindtree Ltd. - Malware Analyst and Threat Researcher

Aug 2023 - May 2025

- Designed and automated a large-scale malware analysis pipeline processing **5K+ executables** daily using **machine learning-based** static analysis, **reducing false positives** by 60% and **triage time by 40%**.
- Built modular **microservice-based** threat classification algorithms for PE, and PUA/UWS files, automated rule generation and enhanced threat mapping accuracy, integrating with Microsoft Defender to boost real-time coverage and reduce manual updates by 25%.
- Performed **reverse engineering**, **debugging**, and **root-cause analysis** on 300+ .NET/MSIL binaries using x64dbg, OllyDbg, DnSpy and Wireshark to identify obfuscation and network-level evasion pattern enabling **zero-day signatures** aligned with MITRE ATT&CK.

LTIMindtree Ltd. - Software Developer Intern

Feb 2023 - May 2023

- Developed and deployed two **cloud-native**, full-stack enterprise applications (Digital Banking, Claims) using Angular, Java, and REST / GraphQL APIs, achieving 99% uptime for **25K+ concurrent users** through **Dockerized microservices** orchestrated with Kubernetes.
- Led feature design, debugging, and performance optimization in **Agile sprints**; implemented **CI/CD pipelines** that accelerated review cycles by 30% and improved release reliability. Awarded **“Star Performer”** for driving **cross-team delivery excellence**.

TECHNICAL PROJECTS

Pack and Verify (Multimodal RAG, Retrieval, Reranking, Verification)

- Built Pack-and-Verify, a **token-budgeted multimodal RAG** system for academic PDFs with **diversity-aware evidence packing** (cosine redundancy penalty), a local **ms-marco-MiniLM-L-6-v2** reranker, and verification/abstention logic for grounded answers.
- Achieved **+4.5% QA accuracy** at a 2K-token budget and **40% token compression** (5K vs 8K-token baseline), improving throughput and reducing long-context inference cost while preserving answer quality.

Blockchain-Based E-Voting System Using Hyperledger Fabric

- Developed a decentralized e-voting platform using **Hyperledger Fabric**, **Go Chaincode**, **Docker**, and **REST APIs**, implementing smart contracts for voter authentication, vote validation, duplicate vote prevention, and immutable transaction management.
- Engineered a permissioned blockchain architecture with **distributed consensus**, **fault-tolerant replication**, and **tamper-resistant audit trails**, enabling transparent election management, secure vote verification, and **end-to-end data integrity**.

Real-time Collaborative Application

- Architected **multi-user** fault tolerant **collaborative platform** using WebSockets, Socket.io, and React.js with operational transformation algorithms for conflict resolution, **supporting 100+ concurrent users** with sub-**50ms latency**.
- Implemented **live cursor tracking**, **real-time text synchronization**, and user presence indicators with Node.js backend and Redis state management, showcasing scalable, event-driven design principles achieving **99.9% uptime**.

LEADERSHIP AND IMPACT

- **President & Treasurer, ACM Student Chapter** - Orchestrated technical workshops and networking events for the ACM Student Chapter, engaging **1,000+ participants** and fostering collaboration.
- Technical Lead, (GDSC) – **Mentored 2,000+** students through AI and Cloud workshops, fostering ownership in modern software practices.